

Linked List - 1

In This Lecture

1. Implementation of Linked Lists
2. Insert, Delete, and Iterate in a Linked List
3. Implementation of Doubly Linked List

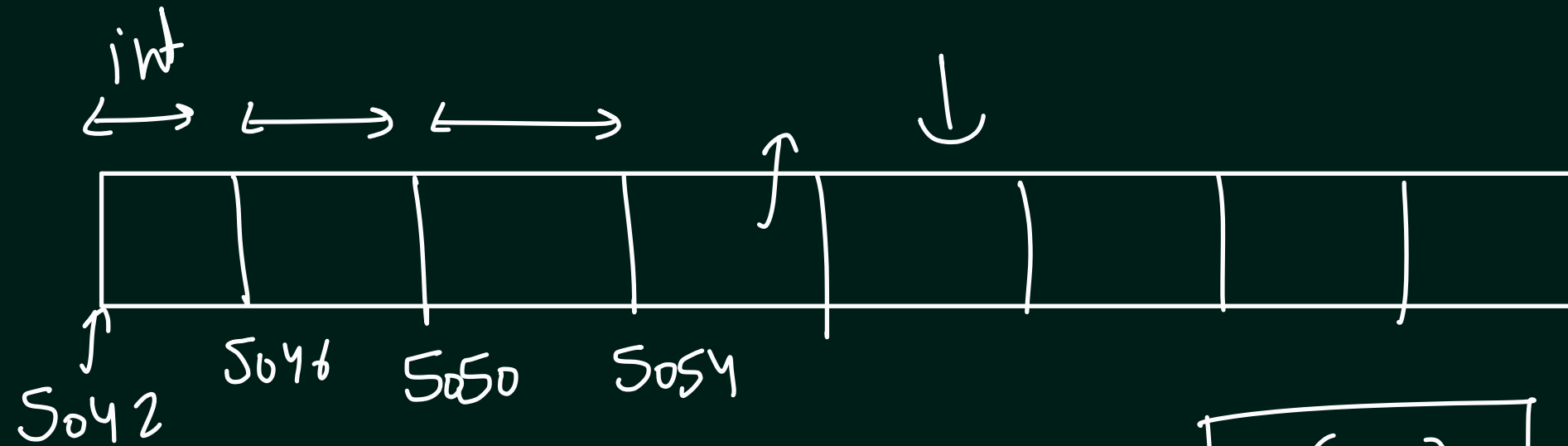
Implementation of Linked Lists

It is a Linear Data Structure where elements are not stored at contiguous memory locations, however as nodes of a LinkedList are connected, it is treated as a Linear Data Structure. A linked list is a collection of nodes where every node contains two fields i.e. data, address fields. The Data field contains the actual value of the node while the address field contains the address of the next node.



Insert, Delete, and Iterate in a Linked List

Contiguous



10^6

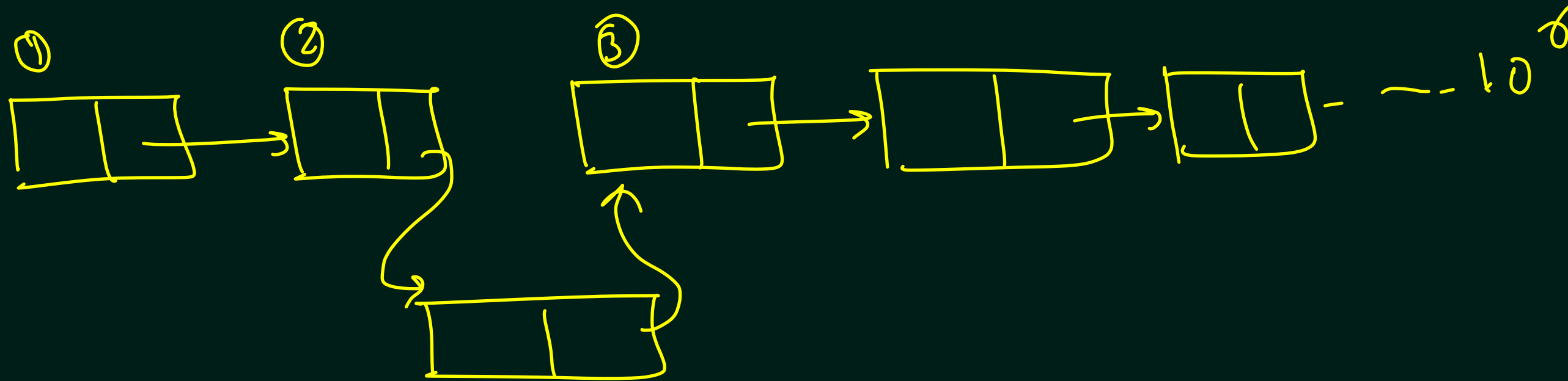
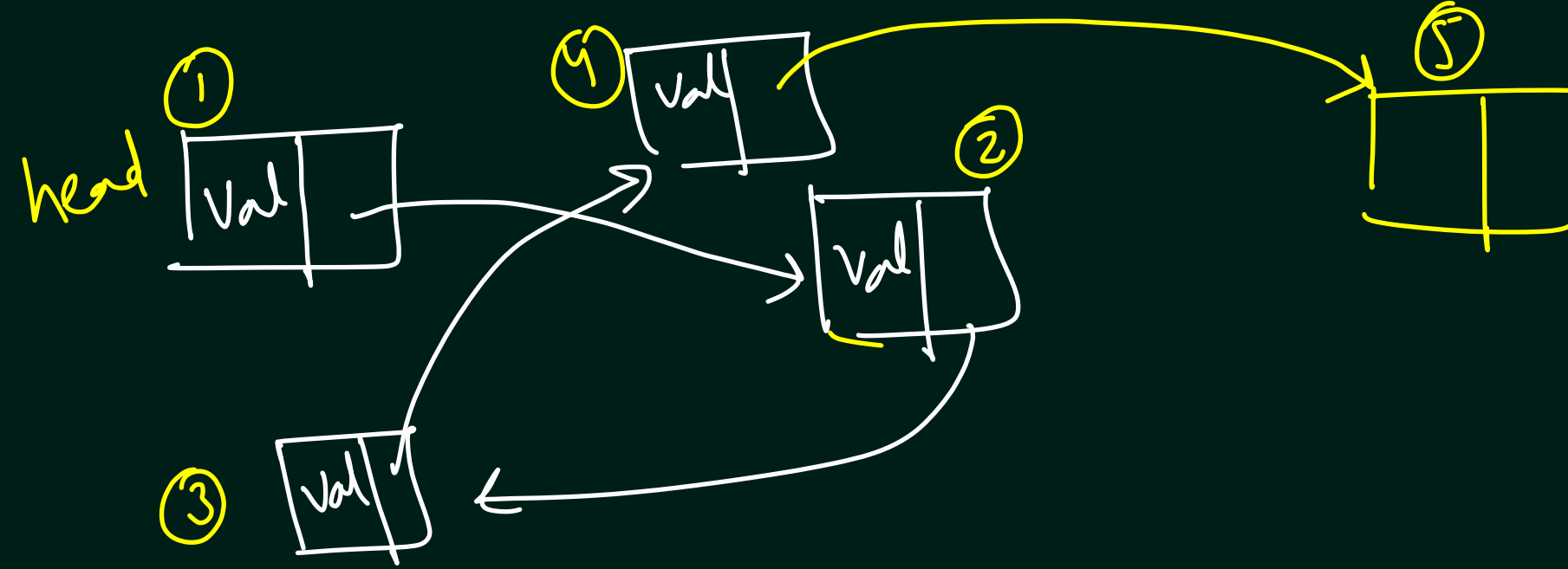
$O(n)$

$$5042 + 3 \times 4$$

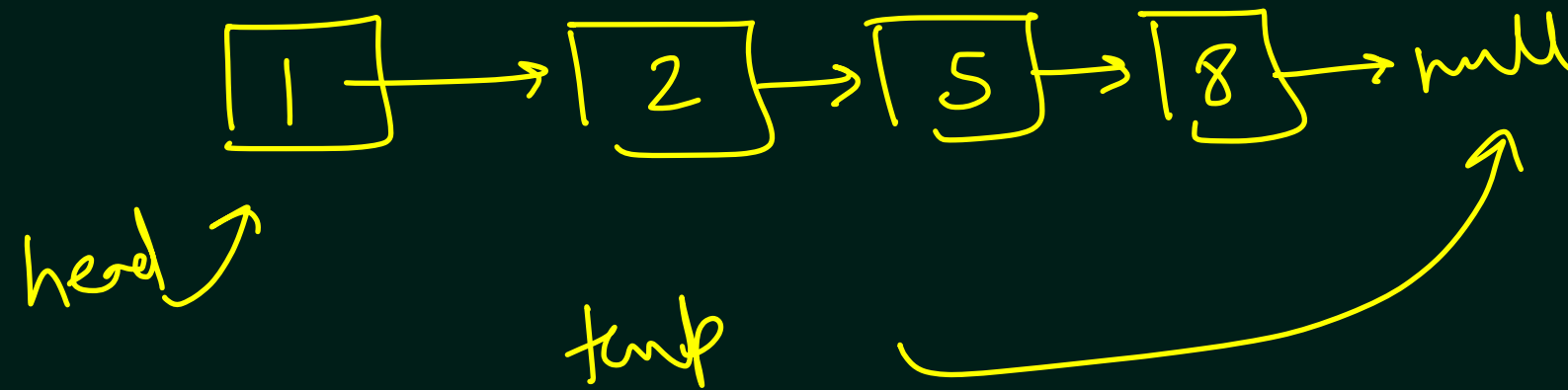
- ① Insert and delete take time
- ② Specify the size before creating array.

Insert, Delete, and Iterate in a Linked List

Non
Contiguous

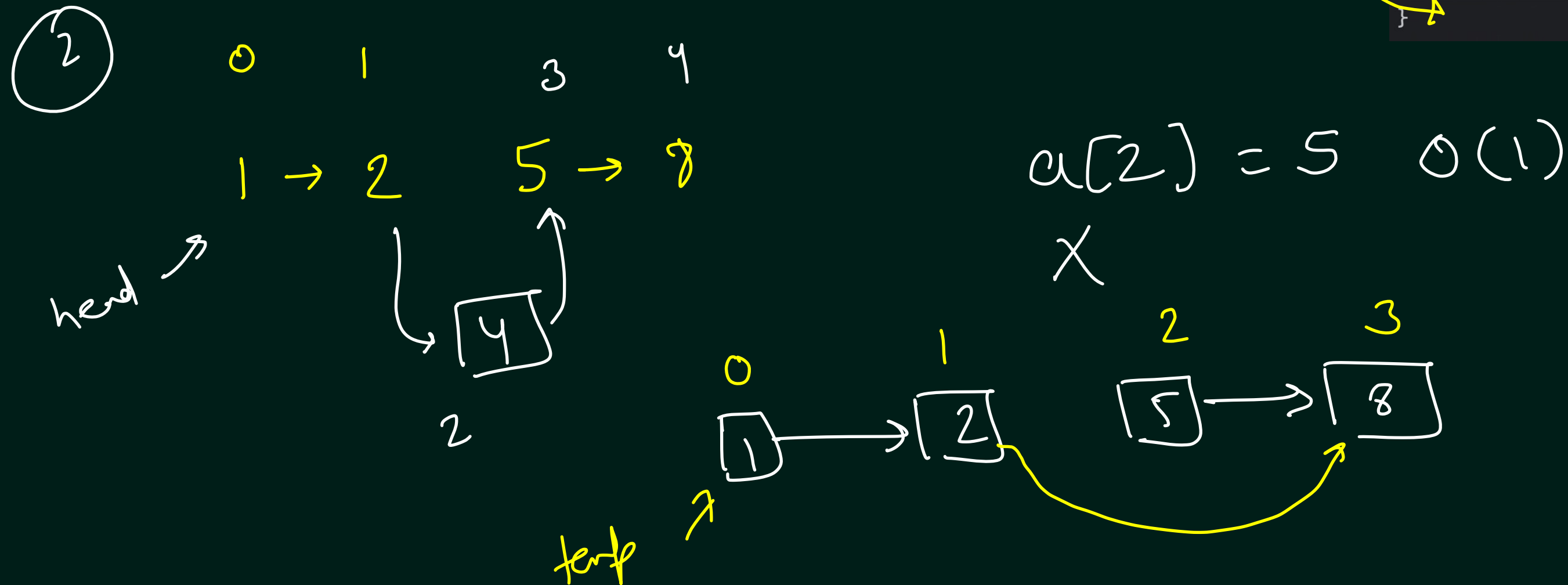


Insert, Delete, and Iterate in a Linked List

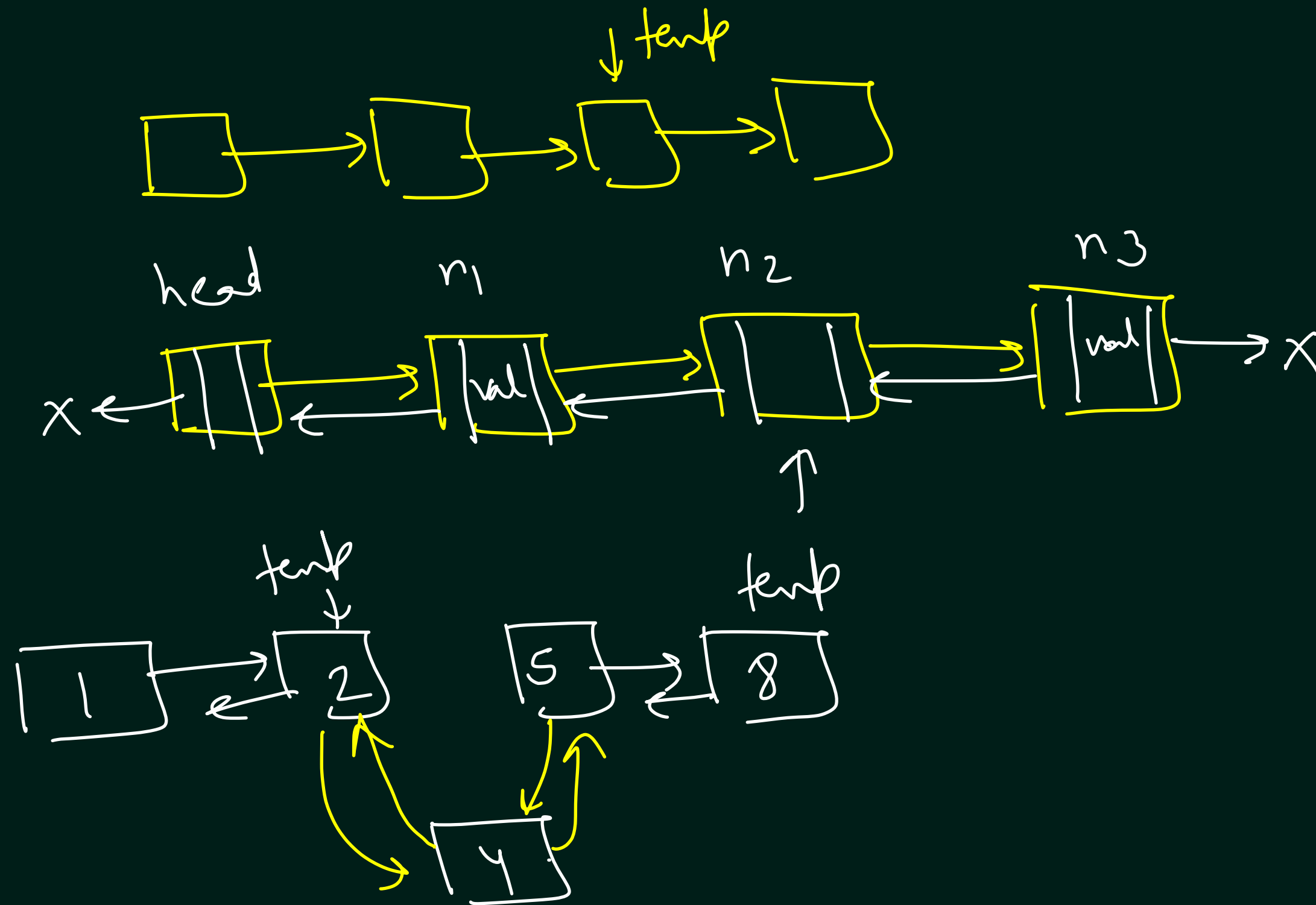


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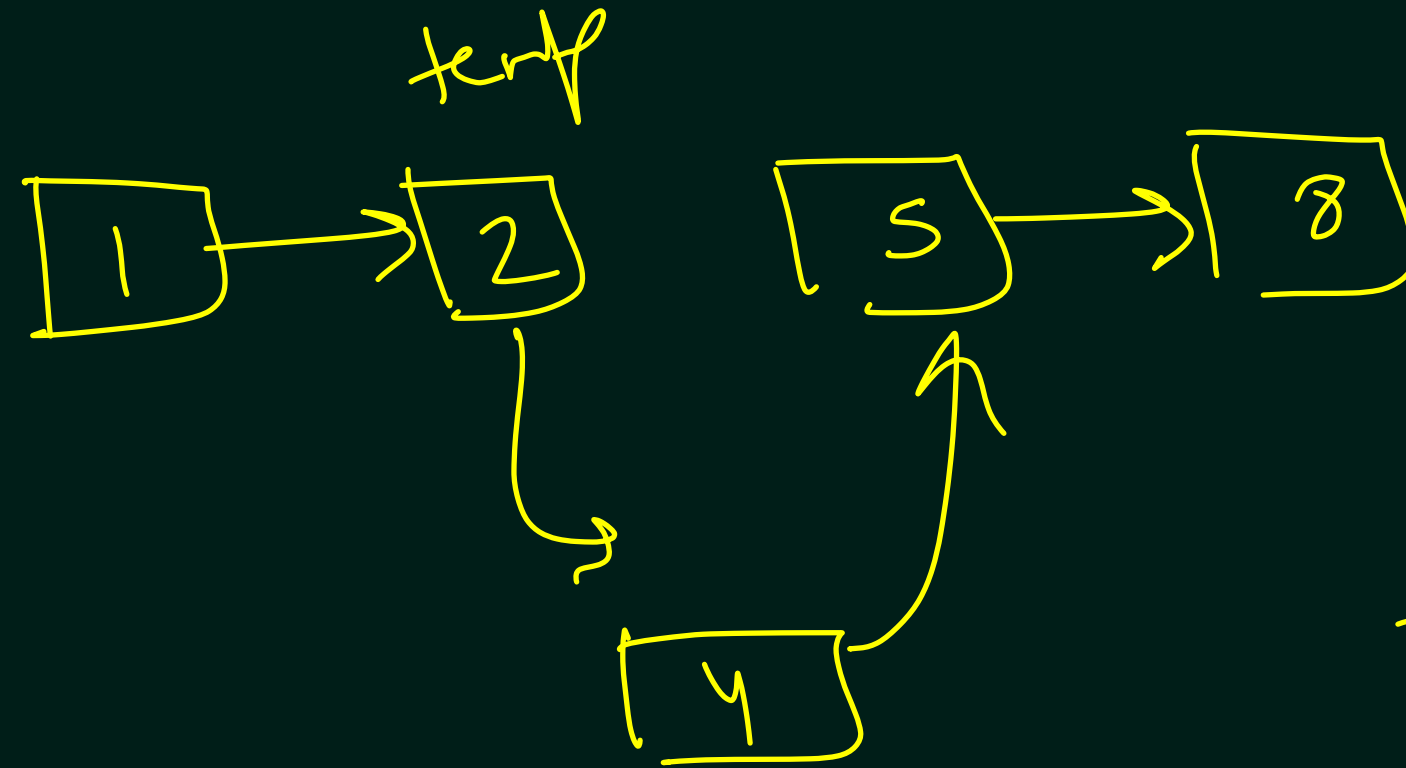
public static void printLinkedList(Node<Integer> head) {
    Node<Integer> temp = head;
    while (temp != null) {
        System.out.print(temp.val + " -> ");
        temp = temp.next;
    }
    System.out.println();
}
  
```



Insert, Delete, and Iterate in a Doubly Linked List



Insert, Delete, and Iterate in a Doubly Linked List



`newNode.next = temp.next`
`temp.next = newNode`